Benchmarking TCR–pMHC structure prediction without multiple sequence alignments: a protein language model matches AlphaFold 3, and their disagreement predicts accuracy

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Abstract

T cell receptor-peptide-major histocompatibility complexes (TCR-pMHCs) are central to the specificity of cellular immunity. With recent breakthroughs in computational methods to model structural interactions, predicting TCR-pMHC structures from sequence has reached new levels of accuracy. Despite these advancements, benchmarking has lagged behind creating uncertainty over the relative performance of these models. Published benchmarks on TCR-pMHC complexes agree that multiple-sequence-alignment (MSA)-based predictors lead the field and that protein-language-models (pLMs) trail them. Here, we revisit that conclusion by evaluating a set of 126 TCR–pMHC complexes(111 class I, 15 class II) on 4 state-of-the-art models (AlphaFold 3, ESMFold2, Protenix, TCRmodel2) covering MSA-based, LM-based, and TCR-specialised architectures. We find that a pLM-based predictor, ESMFold2, matches the top MSA-based predictor, AlphaFold 3, in overall accuracy and surpasses all other models in confidence calibration. (Pearson r = 0.79 vs 0.66); We also find that the two top models differ vastly in their diffusion modules, showing that AlphaFold 3 converges on a structure much later that ESMFold2 and that changing their respective diffusion steps can save computational cost. These results will guide practitioners in selecting the right models and settings for TCR-pMHC prediction tasks.

1 Introduction

T cell are crucial to adaptive immunity by performing various functions involving the recognition of foreign antigens [1,2]. This recognition is mediated by T cell receptors (TCRs) which bind to certain peptide-major histocompatibility complex (pMHC) molecules presented on the surface of target cells [3,4]. The specificity of TCR-pMHC interactions is determined by the structural complementarity between the TCR and the pMHC [5–8]. Thus, understanding the structural basis of TCR-pMHC interactions is essential for various downstream applications such as epitope-specificity prediction, TCR engineering, or immunotherapeutic design [9–11]. Unfortunately, experimental determination of TCR-pMHC complexes [12–14] remains slow relative to the size of the repertoire [15–18], so computational modelling has become the practical route to structural hypotheses at scale.

Deep-learning structure prediction has transformed this problem twice. AlphaFold 2 made substantial improvements in single-chain and many multi-chain predictions [19,20], though early benchmarks identified adaptive immune recognition, inluding TCR–pMHC complexes,as a point of weakness [21]. AlphaFold 3 replaced the structure Module with a diffusion head, substantially improving the interface accuracy across biomolecular space [22], and TCR-specific adaptations such as TCR-dock [23] and TCRmodel2 [24] narrowed the gap for this class of complex specifically. These structure prediction models rely heavily on multiple sequence alignments (MSAs) and structural templates—data inputs derived from previously solved structures that guide the prediction process using coevolutionary information [25]. However, protein language models have made structure prediction possible without relying on MSAs. ESMFold demonstrated that a sufficiently large language model can encode atomic-level structural information within its representations [26], replacing evolutionary coupling signals with a learned prior to achieve an order-of-magnitude speedup. Although ESMFold was accurate for many single-chained proteins, it was far less accurate on complex multi-chain structures, including TCR–pMHC complexes [27,28]. The more recent ESMFold2, a successor to ESMFold, tries to improve on this issue by incorporating a diffusion module as its structure decoder to better sample a more diverse set of geometrically plausible docking configurations [29].

Recent TCR–pMHC benchmarks have converged on a consistent ranking. Recent studies have shown thatMSA-based methods, particularly AlphaFold3, consistently outperform all alternatives, with most PLM-based models underperforming except a PLM variant built on an AlphaFold-3-style architecture [27]. Ascunce-París et al. reached the same conclusion on 265 class I complexes, reporting AlphaFold 3 as the best generaliser to post-training structures [30]. Shi et al. found that TCR-specific tools derived from AlphaFold 2 gain interface accuracy at the cost of framework accuracy, and that CDR3 loops, docking orientation and class II MHC–peptide interfaces remain difficult for every method [28]. Yin et al. observed complementary behaviour between methods and pooling gains for antibody–peptide complexes [31]. Together these establish the field's working assumption: evolutionary information is what makes TCR–pMHC prediction work.

Two considerations motivate re-examining that conclusion. First, the PLM predictors in those comparisons are predominantly ESMFold-generation models whose structure module is a direct AlphaFold-2 descendant, whereas a current-generation language-model predictor with a diffusion structure head is a substantially different architecture. Second, a benchmark that ranks methods addresses treats them as black boxes wherein the internal mechanisms that lead to the result are not considered. With the increase in use of diffusion-based models [32,33], it is becoming easier to evaluate the internal trajectories of these models; following these trajectiories can provide insight the sources of disagreements between models and reveal places to cut computational cost without sacrificing accuracy.

To explore these ideas, we assembled a benchmark of 126 TCR–pMHC post-training cutoff entries. Every method is scored against the experimentally validated biological structure using a variety of metrics including interface accuracy (DockQ) [34,35], TCR-specific accuracy (MHC-aligned weighted all-CDR Cα RMSD and framework-aligned CDR3 all-atom RMSD) [23,36], confidence calibration, sensitivity to reference resolution and to training-set similarity. We then investigate the structural disagreement between the two leading models, AlphaFold 3 and ESMFold2 to understand the sources of their differences. Together, these insights will better guide areas of future improvements to these computational models and assist in practical applications such as immunotherapy and drug design.

2 Materials and methods

2.1 Benchmark set construction

We constructed a dataset of TCR-pMHC Class I and Class II complexes from the TCR3d database [12,14], which curates all known TCR complex structures. Restricting to $\alpha\beta$ TCR complexes with an initial RCSB [37] release date after 30 September 2021(the training cutoff date for all 4 models) gave 427 class I and 72 class II candidates. Of these, 136 were complete TCR–peptide–MHC complexes containing an MHC platform, a peptide, and paired TCR α and β domains (114 class I, 22 class II).

All 136 entries were reviewed against their deposited coordinates and primary publications, and ten were excluded from scoring. Five (8TRL, 8TRQ, 8TRR, 9NIG, 9WNT) contain modified peptide residues that a standard FASTA sequence cannot represent. Four (8VD0, 8VQ8, 9AUD, 9C3E) are engineered constructs in which the peptide is covalently fused to the MHC through a linker. The last, 9J4S, has no physical TCR–pMHC interface meaning it cannot be scored by our standard metrics. (Section 2.3). This leaves 126 scoreable complexes, 111 class I and 15 class II.

No redundancy filter or resolution cut was applied to the primary set. TCR variable-domain identity is a poor novelty axis — germline V-gene framework dominates a ~110-residue V domain — and filtering on it discards most of the set while retaining an idiosyncratic remainder (Section 3.8). Instead, similarity to the pre-cutoff structural record is measured explicitly and treated as a covariate.

2.2 Domain truncation and chain convention

Each complex was truncated to the domains that constitute the variable regions of the TCR–pMHC interface (Figure 1). For class I, β2-microglobulin was removed and the MHC heavy chain was cut to its $\alpha1\alpha2$ peptide-binding platform (median 181 residues, range 161–181); for class II, the $\alpha1$ and $\beta1$ domains were retained (81–85 and 89–93 residues). TCR chains were cut to their variable domains (α 101–114 residues, β 105–116). Prediction inputs used the complete deposited polymer sequences over those spans.

2.4 Structure prediction methods

Four methods were run on identical inputs.

AlphaFold 3 [22] predictions were obtained through the AlphaFold Server, which supplies MSAs and structural templates from its own pipeline and returns five ranked diffusion samples; the top-ranked model by ranking score was scored. The server enforces a 30 September 2021 training cutoff.

Protenix [38], an open reproduction of the AlphaFold 3 architecture, was run through its public server with the same inputs, and the top-ranked model scored.

ESMFold2 [29] was run through the ESM SDK API as model `esmfold2-fast-2026-05` with the SDK defaults of 10 refinement loops and 100 sampling steps. ESMFold2 is a diffusion structure predictor in the same sense as AlphaFold 3, but it takes no MSA and no structural template: its only input is the query sequence, embedded by a 6-billion-parameter protein language model. This is the architectural contrast on which the present comparison rests, and it is a strict one, since the two methods share the diffusion generative head and little else on the conditioning side.

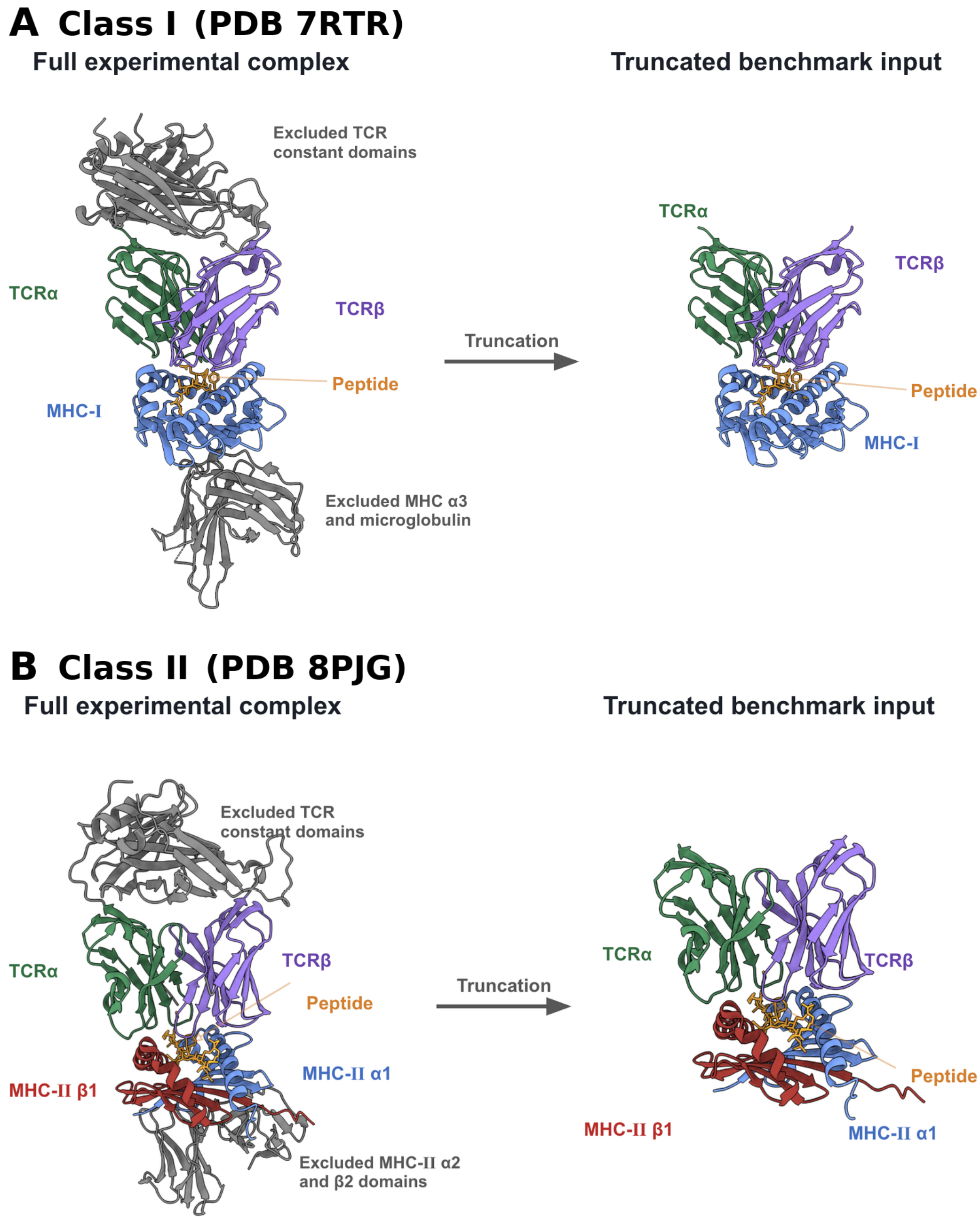

Figure 1: **Domain truncation applied to the benchmark complexes.** (**A**) Class I. The TCR constant domains, the MHC $\alpha 3$ domain and $\beta 2$-microglobulin are removed, leaving the MHC-I $\alpha 1 \alpha 2$ peptide-binding platform, the peptide and the paired TCR variable domains. (**B**) Class II. The TCR constant domains and the MHC-II $\alpha 2$ and $\beta 2$ domains are removed, leaving the $\alpha 1$ and $\beta 1$ peptide-binding domains, the peptide and the paired TCR variable domains. Retained chains are coloured by role; removed regions are grey.

TCRmodel2 [24] was installed locally from the Pierce laboratory repository (commit `90536de`, AlphaFold 2.3 base, `alphafold_params_2022-12-06`) with `--max_template_date=2021-09-30` applied explicitly to both the TCR and the pMHC template searches; the default is effectively unrestricted, so this setting is required for a valid post-cutoff comparison. TCRmodel2 accepts a fixed peptide length for class II and was therefore run on class I only, giving 111 predictions.

Mean end-to-end runtimes were 5.2 s per complex for the ESMFold2 API (median, IQR 4.9–6.4 s) and 13.4 s per complex for a local AlphaFold 3 run at default sampler settings on one RTX PRO 6000 (Section 2.9).

2.5 Accuracy metrics

Interface accuracy was measured with DockQ v2.1.3 [34,35], run once per complex with an explicit identity chain mapping. Without one, DockQ optimises chain assignment and can swap the homologous TCRα and TCRβ chains, silently inflating scores. The primary reported quantity is DockQ v2's Global DockQ, its interface-size weighted score over the whole complex; per-pairwise-interface DockQ values are also retained, giving 6 interfaces per class I complex and 10 per class II complex. Standard CAPRI thresholds are used throughout (< 0.23 incorrect, 0.23–0.49 acceptable, 0.49–0.80 medium, ≥ 0.80 high) [39].

Two TCR-specific metrics were computed on the same top-ranked predictions the DockQ workflow scores; predictions were never selected using a native RMSD. TCRα and TCRβ chains were annotated once per biological sequence with ANARCI [40,41] under IMGT numbering [42], and the annotation propagated to prediction and native coordinates through pairwise sequence alignment rather than through PDB residue numbering. CDR ranges were the inclusive IMGT intervals 27–38 (CDR1), 56–65 (CDR2), 81–86 (CDR2.5) and 104–118 (CDR3); "all CDRs" means all four regions on both chains, eight in total.

The first, the MHC-aligned weighted all-CDR Cα RMSD, follows the TCRdock formulation [23,36]. MHC residues are matched between prediction and native by sequence alignment; a single ordinary-least-squares Kabsch superposition [43] of the class I peptide-binding platform (or, jointly, of both class II peptide-binding domains) is applied to the entire predicted complex, and no further fit is performed. Cα RMSD is then computed over the eight CDR regions with CDR3 residues weighted 3 and all others weighted 1:

$$\mathrm{RMSD}_w = \sqrt{\frac{\sum_i w_i d_i^2}{\sum_i w_i}}$$

Because there is no second fit, this metric measures how accurately the TCR is docked on the pMHC.

The second, the framework-aligned CDR3 all-atom RMSD, measures the opposite quantity. For each chain independently, the predicted variable-domain framework (the IMGT-numbered variable domain excluding all four CDR regions) is superimposed on the native framework using matched non-hydrogen atoms, and all-heavy-atom RMSD is then computed over the corresponding CDR3 loop without further alignment, comparing only identically named atoms in sequence-matched, identity-matched residues. Cα-only versions are retained as quality-control columns.

Every metric reports expected, matched and resolved residue counts, atom counts, coverage fraction and a pass/fail status; incomplete loops are kept separate from the complete-case analysis rather than silently scored. Validation tests confirm that a native scored against itself, and against a rigidly rotated and translated copy of itself, returns approximately zero; that displacing an intact TCR relative to a fixed MHC raises the MHC-aligned CDR RMSD while leaving framework-aligned CDR3 RMSDs near zero; that deforming a CDR3 loop raises its own framework-aligned RMSD; and that permuting chain identifiers changes nothing, because chain assignment is by sequence and biological role.

2.6 Confidence and reference-free quality scores

Each method's own interface confidence (ipTM, and pTM and mean pLDDT where available) [19,20] was taken from its output. pDockQ [44] was computed per interface from the predicted structure and per-residue confidence, and averaged per complex. A published random-forest quality-tier classifier for TCR–pMHC models [30] was re-implemented against this benchmark's chain convention and evaluated on the subset of class I targets whose PDB identifiers do not appear in its training set (n = 27), alongside a composite score combining the weakest pairwise ipTM, pDockQ, CDR3α pLDDT and the worst pairwise PAE from a single AlphaFold 3 prediction.

2.7 Similarity to the pre-cutoff structural record

Because neither AlphaFold 3's nor Protenix's actual training list is public, we used structures available in the PDB on or before 30 September 2021 as a reproducible proxy. Candidate reference entries were the union of TCR3d's class I and class II complex tables and the STCRDab all-structures summary [13] filtered to `TCRtype = abTCR` and `mhc_type` $\in$ `{MH1, MH2}`; of 395 candidates, 259 were released on or before the cutoff per RCSB `initial_release_date`, yielding 1,112 consistently cropped chains from 258 complete complexes. Every polymer entity of every reference entry was aligned by blastp [45] against the benchmark's own cropped chains, and the best-scoring benchmark role defined both the role assignment and the crop, so reference chains are cut to exactly the domain boundaries the benchmark uses and role assignment never depends on a database's chain-letter column. Where an entry's two TCR entities both matched the same role, the pairing was forced and the weaker preference reassigned.

Identity was reported over the benchmark chain length with $\geq 80\%$ query coverage required. Peptides (4–16 residues) are too short for blastp heuristics and were instead locally aligned (BLOSUM62 [46], gap $-11/-1$) against every reference peptide under the same coverage rule. The *closest complete reference complex* requires every corresponding chain to come from one reference entry, and is the entry maximising mean identity across them. No benchmark entry appears in the reference set; all 136 were released on or after 13 October 2021.

2.8 Cross-model divergence

Because AlphaFold 3 and ESMFold2 predict the same FASTA with the same chain letters and 1..N residue numbering, their predictions correspond residue-for-residue with no alignment step. For each complex we computed Cα RMSD between the two predictions in four interface frames (peptide–TCRα, peptide–TCRβ, MHC–TCRα, MHC–TCRβ; interface residues defined as the union over both predictions of Cα within 10 Å across the interface, matching DockQ's cutoff), the peptide RMSD after superposition on the MHC, and the peptide RMSD after superposition on itself. The rigid-body decomposition places both predictions in the MHC frame and reports the rotation and centre-of-mass offset carrying AlphaFold 3's TCR onto ESMFold2's, with the TCR's own internal RMSD reported for contrast.

2.9 Sampler instrumentation and timing

Both models' official samplers were patched observationally to retain the coordinate state after every production denoising step, changing no arithmetic: AlphaFold 3's `diffusion_head.sample` already computes and discards the per-step positions inside its scan, and the patch keeps them; ESMFold2's `DiffusionStructureHead.sample` holds its state in a local, so its installed source was rewritten at a unique anchor. Parity between instrumented and stock runs under the same seed was verified per model (mean $|\Delta \text{DockQ}| = 0.0008$, maximum 0.0122).

Sampler economy was measured on 30 complexes drawn with a frozen seed from the 126 scoreable set (28 class I, 2 class II, 397–416 residues), run serially on one idle RTX PRO 6000 with identical MSAs and templates taken verbatim from the AlphaFold Server bundles the benchmark's own predictions came from, and the same per-complex seed. Four tracks were run: 200 steps $\times$ 5 samples (default), 160 $\times$ 5, 200 $\times$ 1 and 160 $\times$ 1. All 30 complexes fall in AlphaFold 3's 512-token bucket, so JIT compilation and kernel autotuning happen once per track; both were charged to two warm-up predictions run and discarded before timing. Instrumentation costs ~2% and was run as a separate pass, so the baseline track is stock AlphaFold 3.

2.10 Statistics and software

Paired method comparisons use the Wilcoxon signed-rank test on the complexes both methods predicted. Correlations are Pearson and Spearman with bootstrap confidence intervals (10,000 resamples, fixed seed); regressions use ordinary least squares with HC3 robust standard errors. Software versions: Python 3.13.9 and 3.11.16, DockQ 2.1.3, ANARCI 1.3 (HMMER 3.4), Biopython 1.87/1.81 [47], blastp 2.12.0+, NumPy 2.3.5 [48], SciPy 1.16.3 [49], statsmodels 0.14.5, JAX 0.9.1, transformers 4.57.6, esm 3.3.0.

3 Results

3.1 Composition of the benchmark set

The 126 scoreable complexes comprise 111 class I and 15 class II entries, 112 human and 14 mouse, released between 13 October 2021 and 29 July 2026. One hundred and twenty were determined by X-ray crystallography and six by cryo-EM, with resolutions from 1.54 to 4.44 Å (median 2.56 Å, IQR 2.20–3.00 Å). They span 27 distinct MHC alleles — the most frequent being HLA-A*02 (29), H2-Db (10), HLA-A*11 (10), HLA-B*27 (9), HLA-A*24 (9) and HLA-E (7) — and 79 distinct peptide sequences of 8–15 residues (median 9).

3.2 Overall interface accuracy

Every method produced at least an acceptable model of every complex: Global DockQ never fell below 0.23 for any method on any of the 126 complexes (Figure 2A). Median Global DockQ was

0.751 for AlphaFold 3 (IQR 0.571–0.826), 0.736 for ESMFold2 (0.592–0.840), 0.708 for TCRmodel2 on its 111 class I targets (0.590–0.808) and 0.701 for Protenix (0.530–0.799).

The two leading methods are not distinguishable. Paired over all 126 complexes, AlphaFold 3 minus ESMFold2 gives a mean difference of −0.016 and a median difference of −0.002 (Wilcoxon $p = 0.68$). Both, however, separate from Protenix: AlphaFold 3 +0.039 ($p = 0.045$) and ESMFold2 +0.055 ($p = 0.0009$). TCRmodel2 was not separable from any of the three on its class I subset (all $p > 0.23$).

The CAPRI-class view (Figure 2B) tells the same story with a different emphasis. No method produced an incorrect model. ESMFold2 placed the largest fraction in the medium-or-better band (91.3%: 54.0% medium, 37.3% high), followed by TCRmodel2 (88.3%), AlphaFold 3 (84.9%: 50.0% medium, 34.9% high) and Protenix (79.4%). AlphaFold 3's slightly higher median therefore coexists with a slightly heavier acceptable-only tail than ESMFold2's — the two distributions cross.

This result runs counter to the current consensus [27,30]: a predictor that uses neither an MSA nor a structural template matches AlphaFold 3 on post-cutoff TCR–pMHC complexes.

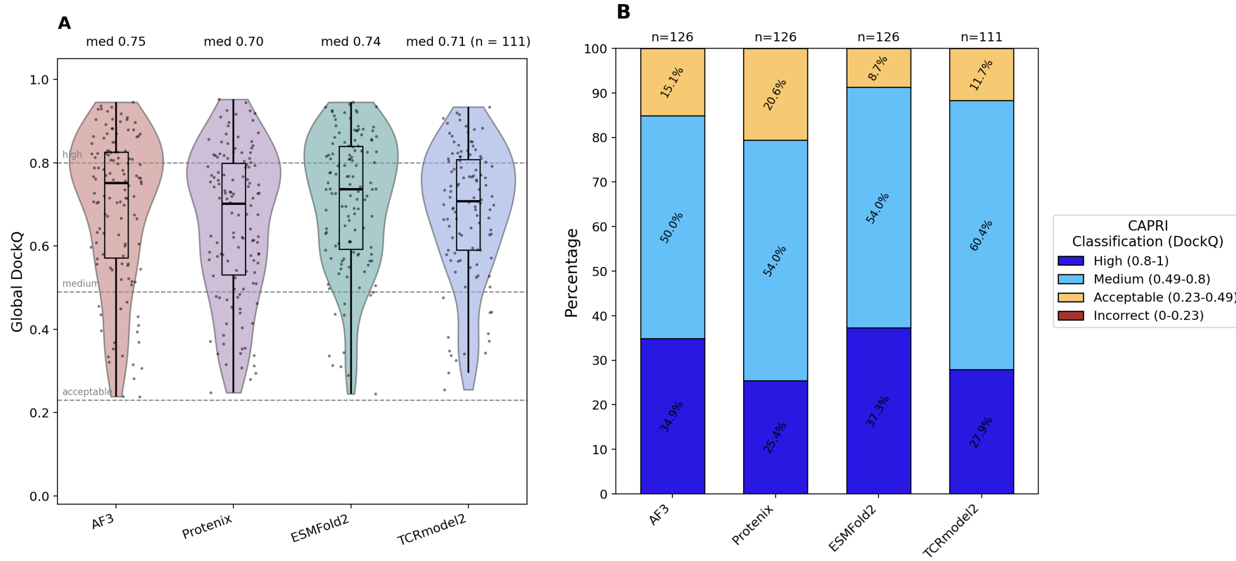

Figure 2: **Interface accuracy on 126 post-cutoff TCR–pMHC complexes.** (**A**) Distribution of Global DockQ per method; box, IQR and median; dots, individual complexes; dashed lines, CAPRI thresholds. TCRmodel2 covers the 111 class I complexes only. (**B**) The same data as CAPRI classes. No method produced an incorrect model of any complex.

3.3 Accuracy by interface

Global DockQ is weighted across all interfaces of the complex, several of which are modelled accurately by every method. Decomposing it by interface (Table 2) localises the differences between methods. The MHC–peptide interface is effectively solved by every method: median DockQ 0.940 (AlphaFold 3), 0.939 (ESMFold2), 0.933 (Protenix) and 0.931 (TCRmodel2), with the class II MHCα–peptide and peptide–MHCβ interfaces close behind (0.90–0.91). The TCRα–TCRβ pairing is also reliable (0.79–0.82). Everything that separates methods, and everything that separates easy complexes from hard ones, is in the four TCR-versus-pMHC interfaces, whose medians run

0.55–0.71. One class II interface is a partial exception: AlphaFold 3 models the MHC-IIα–MHC-IIβ heterodimer pairing itself at median DockQ 0.628, against 0.845 for ESMFold2 and 0.789 for Protenix, the single largest per-interface deficit for AlphaFold 3 in the table.

Table 1: **Median per-interface DockQ by method.** n is the number of complexes contributing that interface (an interface with no native contacts is not scored, so peptide–TCR interfaces are absent in a few complexes whose TCR does not touch the peptide). The final row collapses the four TCR-versus-pMHC interfaces to one value per complex, then takes the median over complexes.

Interface	n	AF3	Protenix	ESMFold2	TCRmodel2
MHC-I – peptide	111	0.940	0.933	0.939	0.931
MHC-IIα – peptide	15	0.904	0.911	0.905	—
peptide – MHC-IIβ	15	0.909	0.887	0.902	—
TCRα – TCRβ	126	0.798	0.800	0.821	0.791
MHC-IIα – MHC-IIβ	15	0.628	0.789	0.845	—
MHC-I – TCRα	111	0.672	0.590	0.653	0.644
MHC-I – TCRβ	110	0.686	0.551	0.696	0.611
peptide – TCRα	120	0.710	0.605	0.676	0.641
peptide – TCRβ	122	0.687	0.633	0.690	0.681
MHC-IIα – TCRα	11	0.615	0.600	0.714	—
MHC-IIα – TCRβ	12	0.773	0.767	0.839	—
MHC-IIβ – TCRα	15	0.720	0.687	0.819	—
MHC-IIβ – TCRβ	15	0.719	0.519	0.764	—
All TCR–pMHC interfaces (per complex)	126	**0.692**	**0.621**	**0.694**	**0.632**

Collapsing those four interfaces to one value per complex, median TCR-versus-pMHC DockQ is 0.694 for ESMFold2, 0.692 for AlphaFold 3, 0.632 for TCRmodel2 and 0.621 for Protenix. On this stricter reading the acceptable-or-better rate falls from 100% to 92.9% (ESMFold2), 90.1% (TCRmodel2), 87.3% (AlphaFold 3) and 84.1% (Protenix), and the high-accuracy rate is 28.6%, 27.8%, 19.8% and 17.5% respectively. Approximately one complex in ten is therefore still docked incorrectly by the best-performing methods.

3.4 TCR-specific accuracy: docking versus CDR3 loop conformation

The two TCR-specific metrics decompose that residual error (Table 3).

Table 2: **TCR-specific structural accuracy, median (IQR) in ångströms, class I and II pooled.** The MHC-aligned metric applies one rigid superposition of the predicted MHC peptide-binding platform onto the native and then measures all eight CDRs with CDR3 residues weighted 3, so it reports docking accuracy. The framework-aligned metrics fit each TCR chain's variable-domain framework independently and then measure the CDR3 loop, so they report loop accuracy independent of docking. TCRmodel2 is scored on the 72 class I predictions that passed CDR annotation and coverage checks.

Metric	AF3	Protenix	ESMFold2	TCRmodel2
MHC-aligned weighted all-CDR	2.84	3.82	2.75	3.53
Cα RMSD	(1.61-5.56)	(2.10-7.47)	(1.60-5.24)	(2.32-4.71)
Framework-aligned CDR3α	1.94	2.33	1.81	1.97
all-atom RMSD	(1.35-2.88)	(1.53-3.08)	(1.25-2.59)	(1.48-2.96)
Framework-aligned CDR3β	1.73	1.93	1.56	1.80
all-atom RMSD	(1.17-2.49)	(1.40-2.60)	(1.13-2.26)	(1.42-2.61)
n complexes	126	126	126	72

The MHC-aligned weighted all-CDR Cα RMSD, which measures docking with no second fit, has a median of 2.75 Å for ESMFold2, 2.84 Å for AlphaFold 3 and 3.82 Å for Protenix over the 126 complexes, and 3.53 Å for TCRmodel2 over the 72 class I predictions that passed annotation and coverage checks. The distributions are strongly right-skewed — means are 5.45, 6.34, 4.83 and 7.72 Å and maxima exceed 50 Å — confirming that the aggregate is dominated by a minority of complexes in which the TCR is placed in an entirely wrong orientation, not by a uniform degradation.

Framework-aligned CDR3 RMSD, which removes docking error by construction, is far more uniform across methods. Median CDR3α all-heavy-atom RMSD is 1.81 Å (ESMFold2), 1.94 Å (AlphaFold 3), 1.97 Å (TCRmodel2) and 2.33 Å (Protenix); median CDR3β is 1.56, 1.73, 1.80 and 1.93 Å. The spread across methods for loop conformation (0.5 Å) is an order of magnitude smaller than the spread across complexes for docking (tens of ångströms), and the maxima are bounded — no method exceeds 7.9 Å on CDR3α or CDR3β. In other words, all four methods build approximately the same quality of CDR3 loop, and differ almost entirely in where they put it. This is consistent with Shi et al.'s observation that CDR3 loops and docking orientation are separate, and separately unsolved, problems [28].

Two secondary observations follow from Table 3. First, ESMFold2's advantage is concentrated in the CDR3β loop and in class II: its class II medians are 1.76 Å (all-CDR), 1.82 Å (CDR3α) and 1.34 Å (CDR3β), against AlphaFold 3's 2.37, 2.37 and 2.05 Å. Second, TCRmodel2's class I all-CDR distribution is the *tightest* of the four (mean 4.83 Å, SD 5.17 Å, maximum 24.4 Å, against AlphaFold 3's 6.18, 9.31 and 52.7 Å): the TCR-specific model is worse on average but fails less catastrophically, which is what one expects from a method whose template search is restricted to TCR and MHC structures.

3.5 Effect of MHC class

Class II is not harder than class I on this benchmark (Figure 3C). Median Global DockQ for class II versus class I is 0.752 versus 0.750 (AlphaFold 3), 0.742 versus 0.736 (ESMFold2) and 0.713 versus 0.698 (Protenix); every difference is well inside the spread, and the class II subset is small ($n = 15$). ESMFold2's class II mean (0.743) is its highest subgroup mean. We note this mainly as a caution: with 15 class II complexes, this benchmark cannot resolve a class effect of the size other studies report [27,28], and the TCRmodel2 comparison is class I only by construction.

3.6 Effect of reference resolution

Accuracy declines with worsening reference resolution for every method, but not equally (Figure 3B). Pearson r between resolution and Global DockQ is −0.292 for AlphaFold 3 ($p = 0.001$), −0.266 for ESMFold2 ($p = 0.003$), −0.238 for TCRmodel2 ($p = 0.012$) and −0.184 for Protenix ($p = 0.040$, Spearman $p = 0.065$, not significant). Part of this is unavoidable — a lower-resolution reference is a noisier target — but the ordering is informative: the two methods most sensitive to reference quality are the two most accurate ones, which is what one expects if the residual error in the best models is approaching the coordinate uncertainty of the references themselves.

3.7 Confidence calibration and reference-free quality scores

Interface confidence predicts accuracy for all three general methods, but ESMFold2's ipTM does so substantially better (Figure 3A). Pearson r between ipTM and Global DockQ is 0.793 for ESMFold2 (Spearman $\rho = 0.777$), against 0.660 for AlphaFold 3 ($\rho = 0.654$) and 0.657 for Protenix ($\rho = 0.679$). The regression lines also differ in intercept: ESMFold2 reaches a given DockQ at a systematically lower ipTM than the other two, so its confidence is both more conservative and better ordered. pDockQ shows the same ranking with lower correlations overall (ESMFold2 $r = 0.676$, Protenix 0.611, AlphaFold 3 0.577).

For AlphaFold 3 specifically, single global confidence scores leave accuracy on the table. Replacing ipTM with the weakest *pairwise* ipTM raises Pearson r from 0.660 to 0.695 over all 126 complexes, and a four-term composite of the weakest pairwise ipTM, pDockQ, CDR3α pLDDT and the worst pairwise PAE — computed from a single AlphaFold 3 prediction, with no second structure and no reference — reaches $r = 0.712$ (95% CI 0.617–0.791) and $\rho = 0.725$ (0.624–0.803). On the same 126 complexes a published random-forest quality-tier classifier [30] reaches $r = 0.662$ and $\rho = 0.674$; the paired difference is +0.050 Pearson (95% bootstrap CI +0.015 to +0.085) and +0.051 Spearman (+0.008 to +0.098). On the 27 class I targets whose PDB identifiers are absent from that classifier's training set, the classifier and ipTM are statistically indistinguishable ($\rho = 0.770$ [0.485, 0.925] versus 0.678 [0.360, 0.874]; paired difference +0.092, CI −0.041 to +0.263). We therefore do not claim a resolved advantage for any single-model score on unseen targets — but we do note that all of these scores are beaten by a quantity that requires no learned quality model at all (Section 3.9).

3.8 Effect of similarity to pre-cutoff structures

Accuracy rises weakly but significantly with similarity to the pre-cutoff structural record. Pooled over 245 predictions, the strongest single predictor is not any individual chain but the composite

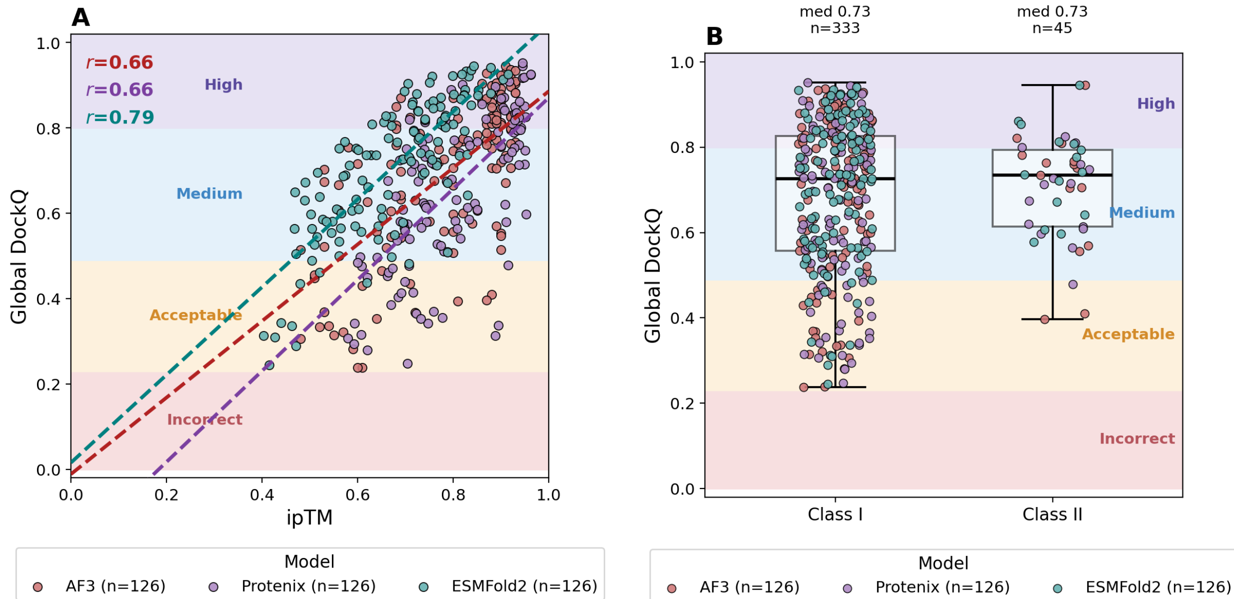

Figure 3: **Determinants of interface accuracy.** (**A**) Global DockQ against each method's own ipTM, with per-method linear fits and Pearson r. ESMFold2's confidence is both more conservative — it reaches a given DockQ at a lower ipTM — and better ordered. (**B**) Global DockQ against the resolution of the experimental reference. (**C**) Global DockQ by MHC class. Shaded bands are the CAPRI quality classes.

mean identity over the closest *complete* pre-cutoff complex (Spearman $\rho = +0.271$, $p = 1.8 \times 10^{-5}$); the individual components are weaker (TCRβ +0.192, peptide +0.151, TCRα +0.127, MHC +0.059, n.s.). The full pre-specified model (DockQ ~ method + TCRα + TCRβ + peptide + MHC + class) explains 7.2% of the variance (adjusted $R^2 = 0.048$), with TCRβ identity worth +0.041 DockQ per 10 identity points ($p = 0.024$) and peptide identity +0.015 ($p = 0.005$). Between the extreme bins of the composite score, mean DockQ rises from 0.515 to 0.702 (AlphaFold 3) and 0.438 to 0.661 (Protenix).

The negative MHC coefficient in that model is an artefact of no variance rather than evidence that MHC familiarity hurts: MHC identity has median 100% and IQR 99.4–100%, because 111 of 126 targets are class I HLA alleles whose peptide-binding domain is in the PDB many times over. A useful counterexample: two targets have a 100%-identical complete pre-cutoff complex — 7RTR (identical to 7N1F) and 9NDS (identical to 4FTV) — and AlphaFold 3 still scores 7RTR at only 0.36, CAPRI acceptable. Sequence-level availability of the answer does not guarantee a good prediction.

The threshold sweep matters for how method comparisons should be read. Filtering at $\leq 95\%$ maximum TCR identity retains 69 targets and barely moves absolute accuracy (AlphaFold 3 mean DockQ 0.585 → 0.547; Protenix 0.515 → 0.506), but it dissolves the AlphaFold 3–over–Protenix result: the paired advantage falls from +0.063 ($p = 0.031$) to +0.027 ($p = 0.40$), and at $\leq 90\%$ it reverses sign. Below 95% the retained samples (14, 7 and 1 targets) are too small to conclude anything either way. We therefore conclude that the AlphaFold 3 advantage over Protenix on this benchmark is carried by targets with a near-duplicate TCR in the pre-cutoff PDB. It is also the reason we did not build the primary benchmark on a TCR-identity filter: only 7 of 126 targets fall below 70% identity, and those 7 are not "hard targets" but mostly unusual or non-human TCRs.

3.9 Structural divergence between AlphaFold 3 and ESMFold2

Equal accuracy is not the same as equal predictions. Across the 126 complexes the median Cα RMSD between the AlphaFold 3 and ESMFold2 predictions of the same FASTA is 1.35 Å, but the distribution runs to 13.9 Å, and the disagreement is anatomically specific. The two models agree closely on everything except where the TCR sits (Figure 4B): median peptide RMSD in the MHC frame is 0.44 Å and median peptide conformation RMSD is 0.33 Å, and the median internal RMSD of the TCR module itself is 0.74 Å — but the median MHC–TCRα and MHC–TCRβ interface RMSDs are 1.62 and 1.63 Å, and the rigid-body decomposition gives a median TCR rotation of 11.1° with a tail reaching 177°. On 20 of 126 complexes the two models place the TCR more than 30° apart, and on 10 of them more than 90° apart, while building the same pMHC and the same TCR fold. Binned, the contrast is stark: the two models place the peptide within 1 Å on 84% of complexes and build the TCR variable module to within 1 Å on 75%, but agree on the MHC–TCRα interface to within 1 Å on only 35% and differ by 2 Å or more on 40% of complexes (11% by more than 5 Å). The disagreement is a docking disagreement, not a folding one.

That disagreement is a strong, entirely reference-free predictor of accuracy (Figure 4A). Cross-model Cα RMSD correlates with AlphaFold 3's Global DockQ at Spearman $\rho = -0.756$ and with ESMFold2's at $\rho = -0.670$, and with the *worse* of the two at $\rho = -0.884$ ($p = 1.1 \times 10^{-42}$). For comparison, on the same 126 complexes AlphaFold 3's own ipTM predicts that worse-of-two value at $\rho = +0.604$ and ESMFold2's at $\rho = +0.639$. Cross-architecture agreement is therefore a better native-free quality signal than either model's internal confidence, than pDockQ, and than the composite and random-forest scores of Section 3.7 — and it requires no confidence head, no training, and no calibration.

By divergence quartile, median Global DockQ falls from 0.860 (AlphaFold 3) and 0.862 (ESMFold2) in the most-agreeing quartile to 0.518 and 0.564 in the most-divergent one. Notably, the two models mostly fail on the *same* complexes: both quartile trends are parallel, and among the 20 most divergent complexes, 4 are failed by both at the medium threshold while 12 are solved by exactly one.

The residual complementarity is nonetheless substantial (Figure 4C). At DockQ ≥ 0.49, AlphaFold 3 succeeds on 107 of 126 complexes and ESMFold2 on 115; the two together succeed on 119 (103 both, 4 AlphaFold 3 only, 12 ESMFold2 only, 7 neither). Pooling an MSA-based and a language-model-based predictor therefore raises the medium-or-better rate from 85% or 91% to 94% — echoing the pooling gains Yin et al. report for antibody–peptide complexes [31] — and, because disagreement itself flags the failures, the pool comes with a built-in triage signal.

Two related class II entries illustrate the pattern concretely. 7SG1 and 7SG2 are HLA-DQ2 complexes with closely related gluten-derived peptides. On 7SG2 both models are correct (AlphaFold 3 0.706, ESMFold2 0.642) and their predictions differ by 2.72 Å with a 14° TCR rotation. On 7SG1 the models diverge by 3.56 Å with a 26° rotation, and AlphaFold 3 drops to 0.396 while ESMFold2 holds at 0.736. Both models report high confidence on both complexes (AlphaFold 3 ipTM 0.87 and 0.92; ESMFold2 0.81 and 0.83) — the confidence heads do not see the failure, and the disagreement does.

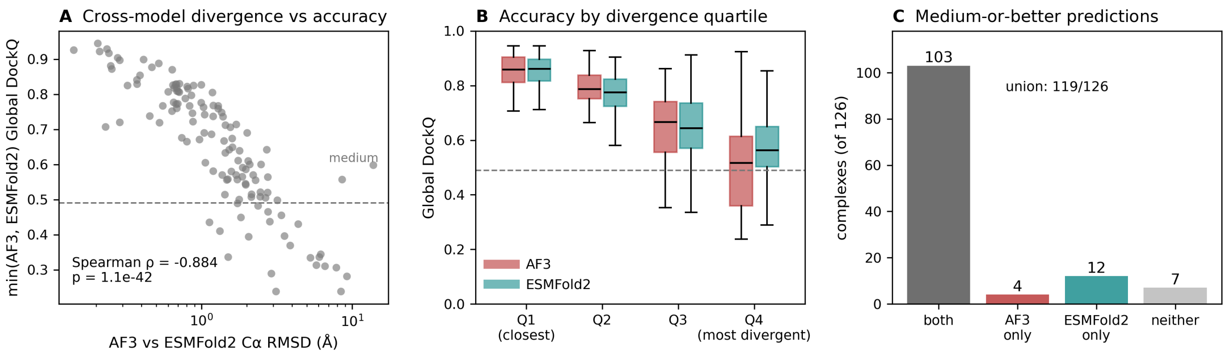

Figure 4: **Structural divergence between the AlphaFold 3 and ESMFold2 predictions.** (**A**) Cα RMSD between the AlphaFold 3 and ESMFold2 predictions of the same sequence against the worse of their two Global DockQ scores. (**B**) The same divergence resolved by region and binned: percentage of the 126 complexes whose two predictions agree to within 1, 1–2, 2–5 or 5 or more Å, with the median beside each bar. The two models build the same peptide and the same TCR fold; they disagree on where the TCR sits. (**C**) Complexes reaching medium-or-better accuracy: individually, jointly, and neither.

3.10 Denoising trajectories of the two samplers

Instrumenting both diffusion samplers on the same complexes shows that the equality of outcome conceals very different dynamics (Figure 5A). AlphaFold 3 begins denoising from an initial noise scale of ~2,560 Å and is still ~44 Å from the native at the midpoint of its 200-step schedule; ESMFold2 begins at ~256 Å and is already within ~5.7 Å at its own midpoint. Measured as the fraction of the schedule at which the trajectory comes permanently within 1 Å of its own final frame, AlphaFold 3 converges at 73% of its steps and ESMFold2 at 58%. AlphaFold 3 spends most of its schedule in a regime where, at the scale the noise schedule sets, no structure has settled at all; ESMFold2 commits early and refines.

The same behaviour at scale (Figure 5B, 30 complexes $\times$ 5 samples = 30,000 recorded frames) puts AlphaFold 3's convergence in a narrow window. Mean Cα RMSD to the final frame of the same sample falls below 2.0 Å at step 150 and below 1.0 Å at step 158; every complex is within 1.0 Å by step 159; accuracy against the native plateaus at 3.37 Å around step 163; and mean RMSD to the final frame falls below 0.5 Å at step 164. Four independent criteria land between steps 150 and 164 of 200. For roughly the first 140 steps the state is noise at the scale the schedule sets — AlphaFold 3's schedule holds σ above 5 Å until ~70% of the way through.

3.11 Effect of AlphaFold 3 sampler settings on accuracy and runtime

If the trajectory has effectively converged by step ~160, the last 40 steps and the four extra diffusion samples should be nearly free to remove. They are (Figure 5C, Table 4). Over the same 30 complexes, same MSAs, same seeds, run serially on one idle GPU:

Table 3: **AlphaFold 3 sampler economy on 30 complexes.** Same complexes, same MSAs and templates, same per-complex seed, run serially on one idle RTX PRO 6000. Confidence intervals are paired bootstrap; p values are paired tests against the baseline track. GPU seconds are mean ± SD over the 30 complexes and exclude a constant 3.3 s of CPU featurisation.

Track	Steps × samples	GPU s per prediction	Speed-up	Mean Global DockQ	Δ vs baseline (95% CI)	Paired p
Baseline	200 × 5	10.04 ± 0.03	1.00×	0.7212	—	—
Fewer steps	160 × 5	8.79 ± 0.03	1.14×	0.7143	−0.007 (−0.026, +0.007)	0.43
One sample	200 × 1	6.02 ± 0.01	1.67×	0.7143	−0.007 (−0.021, +0.006)	0.34
Both	160 × 1	5.57 ± 0.01	**1.80**×	0.7109	−0.010 (−0.032, +0.007)	0.33

Every confidence interval straddles zero, no track loses a complex from the acceptable band (30/30 everywhere), and the high-accuracy fraction moves from 0.367 to 0.333 — one complex. Adding the constant 3.3 s of CPU featurisation gives the end-to-end figure: 13.4 s → 8.9 s per complex, or 401 s → 267 s for all 30. Note that the 160-step track is not the 200-step trajectory stopped early — AlphaFold 3 re-discretises the same σ range into 160 coarser steps — so this is an experiment testing the convergence analysis, not a restatement of it.

The speed-up saturates at 1.8× for a structural reason. Fitting t = trunk + c × (steps × samples) across all 120 timed predictions gives a fixed trunk of 4.86 s and 5.08 ms per sample-step (R^2 = 0.995). Half of a default AlphaFold 3 run on a complex of this size is the Pairformer trunk and its 10 recycles, which neither shortcut touches; driving diffusion cost to zero would give at most 1.9×. Dropping from five samples to one also costs the ranking step — AlphaFold 3 normally returns the best of five by ranking score — and on this sample that selection is worth −0.007 DockQ, which is not detectable at n = 30.

4 Discussion

4.1 Multiple sequence alignments are not required for TCR–pMHC modelling

The clearest result reported here is a negative one: on 126 post-cutoff complexes, a diffusion predictor conditioned only on a protein language model matched AlphaFold 3 running with server-supplied MSAs and templates (median Global DockQ 0.736 vs 0.751, paired p = 0.68), and exceeded it in the fraction of medium-or-better models (91.3% vs 84.9%). This differs from the current consensus [27,30]. The discrepancy appears to be generational rather than contradictory: the PLM predictors in those comparisons are ESMFold-generation models whose structure module descends from AlphaFold 2, whereas ESMFold2 pairs a 6-billion-parameter language model with a diffusion structure head. The most informative data point in Lu et al. is that their best-performing PLM method was the one built on an AlphaFold-3-style architecture [27]; the present result is the same

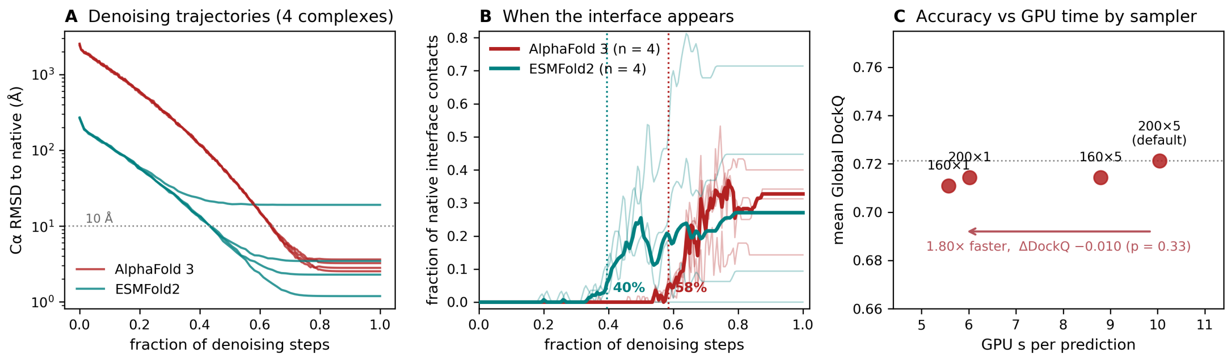

Figure 5: **Denoising trajectories and AlphaFold 3 sampler economy.** (**A**) Cα RMSD to the native along the denoising trajectory for four complexes traced in both models; AlphaFold 3 (200 steps, initial $\sigma \approx 2{,}560$ Å) is still tens of ångströms out at its midpoint, ESMFold2 (68 recorded steps, initial $\sigma \approx 256$ Å) is within a few ångströms. (**B**) AlphaFold 3 convergence to its own final frame over 30 complexes $\times$ 5 samples; band spans median to 90th percentile. (**C**) GPU time against mean Global DockQ for four sampler configurations on the same 30 complexes.

observation with the architecture gap closed further. Why this should hold for this problem in particular is worth stating explicitly: TCR variable domains and MHC platforms are among the most heavily sequenced and most heavily crystallised folds in existence, so an MSA adds little that a large language model has not already internalised, whereas the quantity an MSA cannot supply — the relative orientation of two domains with no coevolutionary history with one another — is precisely what remains unsolved.

4.2 Docking, not loop conformation, is the unsolved problem

The per-interface decomposition and the two CDR metrics point the same way. MHC–peptide interfaces are modelled at DockQ ≈ 0.94 by all four methods; CDR3 loops are built to within 1.6–2.3 Å heavy-atom RMSD of the native by all four after framework alignment, with a spread across methods of half an ångström. Docking accuracy, by contrast, spans tens of ångströms, is right-skewed with catastrophic outliers, and accounts for essentially all of the difference between methods and between complexes. This reproduces the decomposition Shi et al. reported [28] on a larger and more recent set, and it argues that further progress on TCR–pMHC modelling will come from the placement problem, not from loop refinement.

4.3 Cross-architecture agreement as a reference-free quality signal

The Cα RMSD between an AlphaFold 3 and an ESMFold2 prediction of the same sequence predicted the worse of their two DockQ scores at $\rho = -0.884$, against $+0.60$ to $+0.64$ for the models' own ipTM values, ~0.68 for pDockQ, and ~0.67–0.73 for learned quality scores. This is not a subtle effect and it is cheap: one extra prediction from a method that runs in five seconds. It also has a mechanistic reading. Two models that share a generative head but nothing on the conditioning side have largely independent failure modes for the docking problem, so their agreement is evidence in a way that a single model's self-reported confidence — which, as 7SG1 shows, can be high on a wrong answer — is not. A practical consequence follows: for TCR–pMHC modelling campaigns, running both architectures, using their agreement as a triage score, and taking the union of their predictions raises the medium-or-better rate on this benchmark from 85% (AlphaFold 3 alone) to 94%.

4.4 Sampler settings and computational cost

AlphaFold 3's denoising trajectory has converged to within 1 Å of its own answer by step 158 of 200, and running with 160 steps and a single diffusion sample returned the answer in 55% of the GPU time with no detectable accuracy cost over 30 complexes. The scope of that result should be emphasised: n = 30, one GPU, one seed per complex, and one narrow size regime (397–416 residues, all in AlphaFold 3's 512-token bucket). Larger complexes shift the trunk/diffusion split — the trunk scales roughly quadratically in the pair track while diffusion scales with atom count [22] — so the 1.9× ceiling moves. Within this regime, though, the finding is straightforward: for high-throughput TCR–pMHC screening the default settings buy almost nothing, and the compute is better spent on a second architecture, which Section 3.9 shows is worth considerably more than a fifth diffusion sample.

4.5 Implications for benchmark design

Two methodological points generalise beyond this set. First, native construction is not a solved preprocessing step. Forty-six of 136 entries are built from files holding several complete complexes, and the deposited asymmetric unit is often not a complex; six natives required a biological-assembly fetch or a periodic-boundary repair, and one (9J4S) had no physical TCR–pMHC interface in any author-defined assembly, its apparent interface being a lattice contact between two copies. A contact-count check cannot catch this — only requiring all chains to come from a single author-defined assembly does. A benchmark that scores such an entry is scoring noise. Second, redundancy filtering on TCR identity is less informative than it appears for this class of complex: germline V-gene framework dominates a ~110-residue variable domain, so 119 of 126 targets sit above 70% identity to something pre-cutoff, and the $\leq$ 90% subset is 14 mostly non-human or atypical TCRs rather than a representative hard set. The composite identity to the closest *complete* pre-cutoff complex separates the benchmark far better and at usable sample sizes (46 targets have no complete pre-cutoff match at all). We report both, and we report explicitly that the AlphaFold 3–over–Protenix result does not survive a 95% TCR-identity filter, which is the kind of statement benchmark papers should make routinely.

4.6 Limitations

The class II subset (n = 15) is too small to resolve class effects, and TCRmodel2 was run on class I only because it accepts a fixed class II peptide length. Nine entries were excluded for non-standard peptide chemistry or covalent tethering, both of which are biologically interesting categories that sequence-input prediction currently cannot address, so their exclusion reflects a limitation of the methods as much as of the benchmark. "Training set" is a proxy throughout — neither AlphaFold 3's nor Protenix's actual training list is public, and we use PDB availability on or before 30 September 2021 as the closest reproducible stand-in, which cannot account for sequence databases or non-TCR structures the models also saw. The reference set for that analysis is only as complete as TCR3d $\cup$ STCRDab. AlphaFold 3 and Protenix were run through public servers, so their exact data pipelines are outside our control; ESMFold2 was likewise run through the ESM SDK API, and the trajectory analysis of Section 3.10 used the local `biohub/ESMFold2` checkpoint rather than the API model, so its per-frame DockQ values are not interchangeable with the benchmark's. The composite confidence score of Section 3.7 was developed on this benchmark and requires independent validation. Finally, the sampler-economy result is a single-seed, single-GPU, 30-complex experiment in one size regime, as stated above.

5 Conclusions

On 126 TCR–pMHC complexes released after the shared 30 September 2021 training cutoff, a single-sequence protein-language-model diffusion predictor matched AlphaFold 3 and beat an AlphaFold 3 reproduction, with no MSA and no structural template. Both, and a TCR-specialised AlphaFold 2 derivative, build the pMHC and the CDR3 loops to comparable accuracy and differ almost entirely in where they place the TCR. The disagreement between the two architectures is anatomically confined to that placement, and predicts accuracy better than any of the confidence scores evaluated here, so that a two-architecture ensemble is both more accurate (94% versus 85% medium-or-better) and self-triaging. AlphaFold 3's denoising trajectory converges by step 158 of 200, and reducing the sampler to 160 steps and one diffusion sample returns the same answers in 55% of the GPU time. Taken together, these observations favour running two architectures at reduced sampler settings over a single architecture at default settings, and using the agreement between them to identify which predictions can be relied upon.

Data and code availability

The benchmark inputs (136 truncated FASTAs, chain manifests, review records), the rebuilt experimental references and their provenance manifest, all predictions, and the complete scoring, analysis and figure code are contained in the accompanying repository. Per-interface DockQ results are provided in long format (one row per interface plus Global DockQ), TCR CDR RMSD results per prediction with full coverage and quality-control columns, and the sampler-economy tracks with per-complex timings and paired differences. All analyses are reproducible from the recorded commands; software versions and model checkpoint revisions and hashes are recorded in the accompanying manifests.

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